



U-Cker at VBS2026: A Web-Based Interactive Video Retrieval System with Multimodal Query Support

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Abstract. We present U-Cker, a web-based interactive video retrieval system developed for participation in the Video Browser Showdown (VBS) 2026. Building upon our previous version designed for the VR4B session at CBMI 2025, where usability for novice users was emphasized, the current system has been extended to address the broader requirements of VBS. U-Cker supports multimodal query formulation by combining free-text and image inputs with adjustable weights, while a built-in query assistance module supports textual inputs through spelling correction, translation, and reformulation. The retrieval engine uses CLIP embeddings and stores feature vectors of millions of keyframes in GPU memory, enabling real-time similarity computation across large-scale datasets. Compared with the previous version, the system has been adapted to handle the expanded datasets and task types of VBS2026, including AVS, KIS, and VQA tasks, as well as additional video collections beyond the V3C dataset. Preliminary evaluations indicate that U-Cker achieves efficient and accurate retrieval performance under diverse task conditions. These results demonstrate the scalability and flexibility of our system and highlight its potential for competitive performance in VBS2026.

Keywords: Interactive video retrieval · Multimodal search · Query assistance

1 Introduction

The Video Browser Showdown (VBS) is a leading international competition for evaluating interactive video retrieval systems under real-time and time-constrained conditions. Participants are challenged to search within large-scale video collections to solve tasks such as Known-Item Search (KIS), Ad-hoc Video Search (AVS), and Video Question Answering (VQA). In VBS2026, the search collections consist of the V3C dataset [1] as well as domain-specific corpora such as the Marine Video Kit (MVK) [2] and the LapGynLHE dataset for laparoscopic gynecology videos [3].

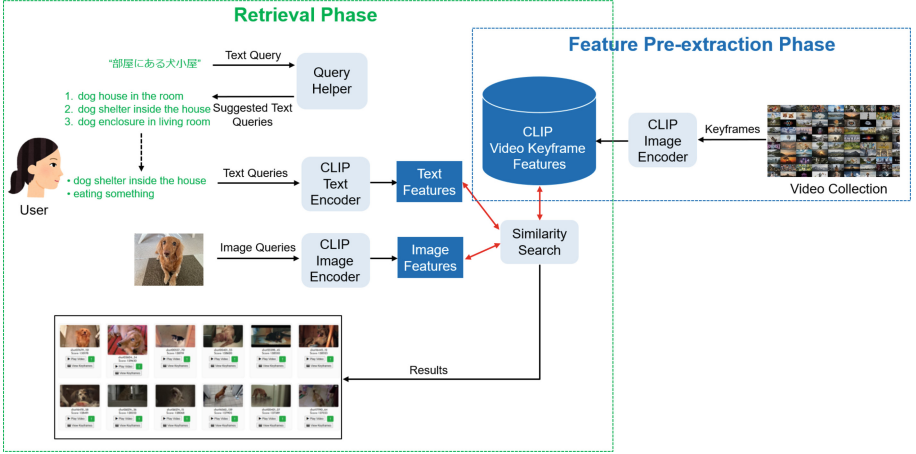


Fig. 1. Overview of the U-Cker system architecture.

Our group has extensive experience from the TRECVID benchmark [4–7], where we focused primarily on accuracy and achieved world-leading performance. However, aspects such as system speed and usability were not emphasized in that context. For the first time, during the Lifelog Search Challenge (LSC’25), we leveraged the engine developed for TRECVID and worked on its acceleration to enable real-time search, which led to the development of the initial version of U-Cker. Subsequently, the system was refined and introduced at CBMI 2025 in the VR4B session, where usability for novice users was emphasized. In this paper, we present its latest version, extended for VBS2026 to integrate multimodal query input, query assistance, and real-time similarity computation using CLIP embeddings [8]. The system is designed to balance usability, scalability, and retrieval performance under the expanded conditions of the competition.

2 System Overview

U-Cker is designed as a fully web-based interactive video retrieval system that emphasizes both scalability and usability. The system runs in a standard browser environment and does not require additional software installation, making it easily accessible for users. The overall workflow of U-Cker is illustrated in Fig. 1. It consists of four main components: feature extraction, multimodal query encoding, similarity computation, and interactive result display. In the feature pre-extraction phase, video keyframes are extracted from the collections and encoded into CLIP feature vectors. In the retrieval phase, users formulate multimodal queries through the input interface. The system subsequently encodes text and image inputs, performs similarity search against the pre-extracted features, and returns ranked results via the result display interface.

At the core of the system lies the retrieval engine, which maps multimodal queries and video keyframes into a shared embedding space using CLIP. For the

U-Cker

Number of images to display: 30

☒ a young girl in the room

Weight: 1.5

Query Helper

☒ eating something

Weight: 1.0

Query Helper

☐ Enter text 3

Weight: 1.0

Query Helper



Cancel

Weight: 1.0

Drop / Click

Cancel

Weight: 1.0

Drop / Click

Cancel

Weight: 1.0

Search

Fig. 2. User interface for composing multimodal search queries. Users can enter text descriptions and reference images, and assign weights to each input.

V3C dataset, feature vectors of over four million keyframes are pre-computed and stored as a dense matrix in GPU memory. This design enables real-time computation of inner-product similarity between queries and all keyframes, without the accuracy degradation inherent in approximate nearest neighbor search. By prioritizing exact similarity computation, the system achieves both precision and responsiveness at the scale required by the competition.

The query input interface supports flexible multimodal query formulation (Fig. 2). Users can combine up to three natural language descriptions and up to three reference images. Each input can be selectively activated and assigned a weight, enabling users to emphasize specific aspects of their search intent. A built-in query assistance module provides spell correction, translation, and reformulation. It remains available to help users refine queries, particularly when English is not their primary language.

The user interface design follows the principles established in the VR4B version of U-Cker, focusing on clarity and minimal cognitive load. The separation between query composition and result browsing is maintained to simplify the workflow. Retrieved keyframes are displayed in ranked order, and interaction options include enlarging an image, playing the corresponding video segment, and browsing adjacent keyframes from the same shot. These functions are

U-Cker Query Helper

Original Query 1

girls eating colorfull ドーナツ in livingrom

Suggested Queries

- girls eating colorful donuts in living room Apply
- colorful donuts in living room with girls Apply
- girls enjoying donuts in living room Apply

Fig. 3. Example interface of the Query Helper. The module provides spell-check, translation, and multiple reformulation suggestions for a given input query.

directly aligned with the needs of VBS tasks such as KIS, AVS, and VQA, where both precise retrieval and contextual exploration are essential.

3 Implementation and Features

3.1 Query Assistance

To support users in composing effective queries, each text input field is linked to a query assistance module based on the Qwen3-8B model¹. When activated, the interface shown in Fig. 3 appears, providing real-time suggestions for refining the user’s input. The visible output consists of multiple reformulated queries, but internally these results are generated by a single structured prompt that simultaneously handles spelling correction, translation, and paraphrasing. This design is efficient, as one compact prompt instruction enables the model to realize three different functionalities without requiring multiple API calls or separate processing stages. The functionality is controlled by a prompt designed to enforce concise and consistent output, for example:

You are a helpful assistant that rewrites noisy or misspelled search queries.
 You should fix spelling mistakes, translate if needed, and output exactly 3 clean,
 short, English search queries suitable for video search.
 Do not include any explanations or comments.
 Return only the rewritten queries as a numbered list.
 Input query: {original_text}
 Output queries:
 1.

¹ <https://huggingface.co/Qwen/Qwen3-8B>.

By using this structured prompting, the model generates real-time suggestions that are particularly beneficial for non-native speakers and for tasks where query intent must be expressed with high precision. By integrating these assistance tools directly into the interface, users can refine their input without interrupting the search workflow.

3.2 Retrieval Engine

The retrieval engine is built upon the OpenCLIP model² [9], which projects both text and images into a shared 768-dimensional multimodal embedding space. For each keyframe, the image encoder extracts a 768-dimensional visual feature vector. In the case of the V3C dataset, which contains the largest number of keyframes among the collections used, these feature vectors are stored in GPU memory as a dense matrix of size $768 \times 4,143,673$.

Similarity scores are computed as inner products between query vectors and all stored keyframe vectors, ensuring exact and deterministic ranking. While approximate nearest neighbor methods such as FAISS or Milvus are often employed for scalability, we prioritize exact computation in order to guarantee reliability under competitive conditions like VBS. The architecture remains scalable to extended datasets, thus maintaining both retrieval accuracy and system responsiveness in real time.

3.3 User Interaction

The system presents the top- k keyframes ranked by the final similarity score. For each retrieved result, users can perform the following interactions:

- Click on an image to enlarge it for detailed inspection.
- Use the [Play Video] button to play the corresponding video segment.
- Use the [View Keyframes] button to browse adjacent keyframes from the same video.

This design enables flexible and efficient multimodal search over millions of video segments while maintaining scalability, and it supports fast decision-making under the time-constrained conditions of VBS tasks.

3.4 Observed Behavior in Preliminary Use

In preliminary tests with example tasks from VBS2025, the system successfully retrieved target scenes in text-based Known-Item Search (KIS-T) when queries contained distinctive cues such as “a wooden cross in front.” Similarly, in video-based Known-Item Search (KIS-V), concise text queries written based on reference clips (e.g., “an old man and a horse”) were effective for locating unique targets. For Ad-hoc Video Search (AVS), combining multiple natural language

² `coca_ViT-L-14` model (pre-trained: `mscoco_finetuned_laion2b_s13b_b90k`).

descriptions and adjusting their weights helped refine complex queries, while for Video Question Answering (VQA), the ability to browse adjacent keyframes was useful for confirming temporal relations.

These observations demonstrate that U-Cker’s core functions, including multimodal input, query assistance, and interactive browsing, are not limited to KIS but are also applicable to broader tasks such as AVS and VQA. They further indicate that the design of GPU-resident feature storage enables real-time retrieval at the scale of more than four million keyframes, which is suitable for interactive competition settings.

In preliminary tests, the query assistance reduced spelling or translation errors and improved retrieval ranking for ambiguous queries. The average latency of query reformulation was below one second on a local GPU server, confirming that the module operates interactively without noticeable delay.

3.5 Dataset Adaptation

While earlier versions of U-Cker focused exclusively on the V3C collection, the system has been extended for VBS2026 to handle additional domains such as underwater diving videos and laparoscopic surgery recordings. Each dataset poses distinct challenges: underwater scenes often feature low contrast and repetitive patterns, while surgical videos contain specialized visual content unfamiliar to general-purpose models. Underwater and surgical videos were embedded using the same OpenCLIP model as the V3C dataset. However, successful retrieval in these specialized domains cannot be fully guaranteed when users lack domain-specific knowledge, as the system relies on general-purpose models. This limitation highlights the need for additional mechanisms to better support such heterogeneous collections.

4 Conclusion and Future Work

We presented the latest version of U-Cker, a web-based interactive video retrieval system developed for VBS 2026. The system integrates multimodal query input, query assistance, and exact similarity computation on millions of keyframes, achieving real-time and accurate retrieval across diverse tasks. Preliminary use confirmed its effectiveness, while also revealing limitations in handling temporally complex events and in adapting to specialized domains such as underwater or surgical videos.

To address these issues, we plan two main extensions. First, we aim to incorporate temporal modeling, enabling the system to handle multi-step or evolving events beyond single keyframes. Second, we will design features that allow effective retrieval in specialized domains even without domain-specific expertise, including techniques such as visual clustering and timeline-based browsing. Through these improvements, we seek to strengthen U-Cker’s performance and adaptability, and to contribute to advancing the state of the art in interactive video retrieval.

References

1. Rossetto, L., Schuldt, H., Awad, G., Butt, A.A.: V3C – a research video collection. In: Kompatsiaris, I., Huet, B., Mezaris, V., Gurrin, C., Cheng, W.H., Vrochidis, S. (eds.) *MultiMedia Modeling. MMM 2019. LNCS*, vol. 11295, pp. 349–360 . Springer, Cham (2019). https://doi.org/10.1007/978-3-030-05710-7_29
2. Truong, Q.T., et al.: Marine Video Kit: A New Marine Video Dataset for Content-Based Analysis and Retrieval. In: Dang-Nguyen, D.T., et al. (eds.) *MultiMedia Modeling. MMM 2023. LNCS*, vol. 13833, pp. 539–550. Springer, Cham (2023). https://doi.org/10.1007/978-3-031-27077-2_42
3. Nasirihaghighi, S., et al.: GynSurg: A Comprehensive Gynecology Laparoscopic Surgery Dataset, [arXiv:2506.11356](https://arxiv.org/abs/2506.11356), 2025
4. Awad, G., Fiscus, J., Godil, A., Diduch, L., Graham, Y., Quénot, G.: TRECVID 2024 - Evaluating video search, captioning, and activity recognition. In: *Proceedings of the TRECVID 2024*, 2024
5. Ueki, K., Suzuki, Y., Takushima, H., Okamoto, H., Tanoue, H., Hori, T.: Waseda_Meisei_SoftBank at TRECVID 2022. In: *Proceedings of the TRECVID 2022*, 2022
6. Ueki, K., et al.: Waseda_Meisei_SoftBank at TRECVID 2023. In: *Proceedings of the TRECVID 2023*, 2023
7. Ueki, K., et al.: Softbank-Meisei at TREC 2024 Ad-hoc Video Search and Video to Text Tasks. In: *Proceedings of the TRECVID 2024*, 2024
8. Radford, A., et al.: Learning Transferable Visual Models From Natural Language Supervision, [arXiv:2103.00020](https://arxiv.org/abs/2103.00020), 2021
9. Cherti, M., et al.: Reproducible scaling laws for contrastive language-image learning. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 2818–2829, 2023